Swati Aggrawal

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PROFESSIONAL SUMMARY

MSc Computer Science graduate from Sapienza University of Rome specializing in Artificial Intelligence and Machine Learning. Author of an IEEE conference publication with research experience in Continual Learning, Transformer Architectures, Computer Vision, and Deep Learning. Experienced in designing experiments, developing and evaluating machine learning models, conducting quantitative analysis, and implementing research pipelines using Python and PyTorch. Interested in representation learning, model interpretability, foundation models, and trustworthy AI.

SKILLS

- **Programming & Databases :** Python, SQL, C++, Bash
- **Machine Learning & AI :** Machine Learning, Deep Learning Architectures, Transformer Architectures, Vision Transformers, PyTorch, TensorFlow, Scikit-Learn, OpenCV, Supervised Learning, Unsupervised Learning, Transfer Learning, Representation Learning, Computer Vision, Generative AI, Classification, Regression, Performance Metrics
- **Data Analytics :** Pandas, NumPy, Matplotlib, Seaborn, Feature Engineering, Data Cleaning, Data Preprocessing, Exploratory Data Analysis (EDA), Predictive Analytics, Model Evaluation, Data Visualization, Performance Benchmarking
- **Research & Evaluation :** Experimental Design, Research Methodology, Literature Review, Performance Benchmarking, Statistical Analysis, Technical Documentation, Scientific Communication
- **Backend & APIs :** FastAPI, REST APIs
- **Tools:** Git, GitHub, Linux, Docker, AWS

EDUCATION

Sapienza Università di Roma **May 2026**

Master of Science in Computer Science

- **Relevant Coursework:** Machine Learning, Computer Vision, Deep Learning and Applied AI, Natural Language Processing, Distributed Systems
- **Thesis:** Evaluating Pretrained Vision Transformers Under Continual Learning.
- Selected for a competitive international master's program.

Sharda University **July 2023**

Bachelor of Technology - CSE with AI and ML

- **Relevant Coursework:** Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- **Thesis:** Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded **merit-based scholarship** (Top 5–10% of cohort).

ML RESEARCH EXPERIENCE

Continual Learning for Computer Vision (Master's Thesis) **Nov 25-May 26**

- Built and evaluated deep learning models using accuracy and comparative benchmarking metrics to identify optimal approaches for sequential learning tasks.
- Designed experimental pipelines for training, evaluation, and comparison of computer vision models across sequential tasks.
- Investigated representation learning, knowledge retention, and catastrophic forgetting through controlled experiments on transformer architectures under continual learning settings.
- Developed reproducible training and evaluation pipelines using Python and PyTorch.
- Automated experiment tracking and performance benchmarking across multiple continual learning scenarios.
- Documented experimental findings and identified research opportunities for future development of continual learning systems.
- Performed statistical analysis of experimental outcomes and interpreted results to support model selection decisions.
- **Technologies:** Python, PyTorch, Vision Transformers, Computer Vision

Machine Learning for Air Pollution [IEEE Publication] **May 2023**

- Published IEEE conference paper evaluating machine learning approaches for environmental prediction and monitoring.
- Analyzed Random Forest, SVM, Genetic Algorithm, LS-SVM, Decision Trees, and Artificial Neural Networks across

multiple air quality datasets.

- Performed comparative evaluation of model performance using statistical metrics to identify the most effective predictive approaches and generate actionable insights.
- Conducted comparative experimental studies across multiple machine learning algorithms and datasets to investigate predictive performance under varying environmental conditions.
- Presented scientific findings through structured research documentation and publication.
- Interpreted experimental results and communicated findings through technical reports and research publication.

PROJECTS

Lossy Text Compressor [Semantic NLP Pipeline]

- Investigated semantic text compression using transformer embeddings, quantization, and reconstruction techniques.
- Evaluated reconstruction quality using cosine similarity, ROUGE scores, and embedding-based semantic similarity metrics.
- Conducted performance benchmarking across multiple quantization levels to analyze trade-offs between compression efficiency and information retention.
- Performed statistical analysis of reconstruction outcomes to identify optimal compression configurations and support data-driven model selection.
- Designed and executed experimental studies to evaluate the impact of quantization levels on semantic information retention.
- Explored representation learning behavior under varying quantization levels to analyze semantic information preservation in transformer embeddings.
- Investigated reconstruction behavior across multiple compression settings using quantitative evaluation metrics.
- Communicated experimental findings and model evaluation results through technical documentation and comparative analysis reports.
- **Technologies:** Python, PyTorch, Transformer Models, Sentence Embeddings

DocDelta AI-Powered Document Intelligence & Comparison Platform

- Developed a document comparison and semantic change detection platform using Sentence Transformers and embedding-based similarity techniques.
- Built REST APIs using FastAPI for automated document analysis and change tracking workflows.
- Performed data preprocessing and feature extraction on document datasets to support semantic comparison and document intelligence tasks.
- Evaluated model performance using similarity metrics and validation techniques to improve semantic matching accuracy.
- Implemented semantic matching to identify added, removed, and modified content across document versions.
- Generated actionable insights from document comparison outputs to support stakeholder decision-making workflows.
- Containerized the application using Docker and deployed it on AWS ECS Fargate with monitoring through AWS CloudWatch.
- **Technologies:** Python, FastAPI, Sentence Transformers, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

- Developed and evaluated computer vision models for automated metal surface defect detection using the NEU Metal Surface Defects dataset.
- Designed a comparative study of CNN, Support Vector Machine (SVM), and Random Forest approaches to analyze the effectiveness of deep learning versus traditional machine learning techniques for industrial inspection.
- Implemented image preprocessing, data augmentation, feature extraction, and model training pipelines using TensorFlow, Keras, NumPy, and Scikit-learn.
- Performed systematic model evaluation using accuracy metrics, confusion matrices, and class-level performance analysis across multiple defect categories.
- Investigated the applicability of computer vision techniques for automated visual inspection, defect classification, and quality-control systems in manufacturing environments.
- Compared deep learning and traditional machine learning approaches to identify trade-offs between accuracy, computational cost, and deployment feasibility.
- Achieved 97.22 % classification accuracy using CNN models and compared performance against SVM and Random Forest baselines.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Scikit-Learn